

Illicit Drug Substitution and Crime Dynamics: Evidence from the Taliban's Opium Ban

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Abstract

Understanding how illicit drug markets and associated social harms respond to exogenous shocks is critical for effective policy formulation. The secretive nature of illicit drug markets makes their study especially difficult. The Taliban's Afghanistan ban on opium production in April 2022 provides a natural experiment for examining drug substitution behaviors and their social consequences. By analyzing international opium price fluctuations resulting from this ban and employing wastewater-based epidemiology to track consumption changes in Australian communities, I identify shifts in drug use patterns. Furthermore, I explore the impact of these market disruptions on community crime rates. The findings offer valuable insights into the adaptive behaviors of drug users amid market shocks and contribute to the evaluation of policy interventions aimed at mitigating harm in illicit drug markets.

Keywords: Illicit Markets, drugs, heroin, crime, supply shocks **JEL Codes:** K42, D12, E26, I18, F14
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1 Introduction

Illegal drug markets are large and highly consequential for society. Estimates from the United Nations Office on Drugs and Crime (UNODC) put the size of the global illegal market at approximately \$XXX billion per year, representing an estimated XX million users across all drug types. Beyond the associated loss of tax revenue for governments, these markets generate substantial negative externalities, including overdoses (cite), adverse long-run health outcomes (cite), and violence (cite). What many policymakers describe in broad terms as “social problems” are, at their core, outcomes of a market in which firms produce and distribute products, incur and respond to costs, and collect revenue while consumers who are constrained by budgets seek to maximize utility from consumption.

Historically, governments have combated illegal drug markets primarily through punitive approaches. Policies targeting these markets have emphasized arresting dealers and users, disrupting trafficking organizations, and interdicting shipments, with the ultimate goal of reducing the supply and availability of targeted substances. The effectiveness of such interventions depends crucially on the price elasticities of users of the targeted drug and on their ability and willingness to substitute to other substances. When demand is inelastic, large supply shocks may raise prices and increase financial strain on users without meaningfully reducing consumption. As outlined in Becker (2006) [2], such conditions can even increase profitability for surviving suppliers that evade enforcement, potentially strengthening remaining criminal organizations and intensifying the harms associated with prohibition.

In this paper, I estimate the consumption responses of drug users to a large, unexpected supply shock. In April 2022, the Taliban in Afghanistan announced a ban on the production and consumption of narcotics within the country. Because Afghanistan has historically supplied the bulk of the global world’s opium, this decree translated into an approximate XX-80% reduction in global opium supply. This shock represents what is likely the largest single drug market intervention in modern history and was followed by a roughly ten-fold increase in the farm-gate price of opium. I use this increase in the international price of opium to measure the severity of the shock, allowing me to estimate the own-price elasticity of demand for heroin as well as cross-price elasticities with respect to other drugs in order to uncover substitution patterns. I then link these estimated elasticities to local crime outcomes, exploiting the timing of the price movements to study effects on different categories of crime.

I find some evidence of reductions in heroin consumption following the price increase induced by the Taliban ban, with lower reductions in consumption from areas with higher pre-ban levels of heroin consumption. I also document substitution into other drugs, most notably fentanyl, suggesting that consumers respond by switching toward more potent and potentially more lethal substances. On the crime side, I find increased rates of petty theft and other economically motivated crime, but no detectable aggregate effect on violent or personally motivated crimes. These results underscore the role of financial pressure in shaping individuals’ propensity to engage in acquisitive crime, and they highlight how supply-side interventions in drug markets can shift rather than resolve underlying harms, both in terms of substance use and criminal activity.

The Taliban’s 2022 opium ban is an unusually natural experiment for studying illegal drug market responses to large shocks. It was substantial in magnitude, largely unanticipated, and has so far been persistent. In many respects, it mimics the idealized objective of aggressive law enforcement or interdiction campaigns where a targeted drug is suddenly and sharply reduced from global supply. Observed market reactions to this ban can therefore be informative about the likely consequences of more conventional enforcement strategies aimed at reducing supply in illegal drug markets.

I contribute to several strands of the literature. First, I add to work that focuses explicitly on price elasticities in illegal drug markets. To understand and predict market responses to enforcement, interdiction, and policy changes, one must know how sensitive drug consumption is to price and how consumers substitute across substances when relative prices change. Gallet (2014) (author?) [5] provides a meta-analysis that synthesizes 42 studies reporting a total of 462 price elasticity estimates across a range of drugs and regions. He documents wide variation in estimated elasticities, including some surprising positive price elasticities, and show that results depend heavily on methodological choices, samples, and outcome measures. Using meta-regressions, Gallet produces an estimate for the price elasticity of heroin of approximately -0.51. He also finds relatively low elasticity of cannabis (-0.15), suggesting cannabis demand is more inelastic than demand for heroin. Overall, the meta-analysis highlights the substantial uncertainty surrounding elasticity estimates in clandestine markets and the difficulty of cleanly identifying demand responses when prices, enforcement, and measurement all move together.

Using exogenous shocks to markets can help researchers to isolate consumers decisions from broader trends happening in the market. A second literature focuses on work that seeks to produce causal estimates of behavior in illicit markets. Reliable data on drug production and consumption are notoriously scarce, and measurement is further complicating by under-reporting, changes in enforcement intensity, and the clandestine nature of the the market itself. Researchers have often relied on proxies for drug use or market intensity and on quasi-experimental shocks to identify effects. A key line of work examines supply shocks induced by regulation of precursor chemicals. For instance (author?) [3] study laws that tightened control over sales of precursor chemicals for methamphetamine and measure impacts across several proxies for use, including hospital admissions and self-reported treatment admissions. They find modest negative effects on methamphetamine use. Similarly, (author?) [4] exploit the same 1995 U.S. government intervention in pseudoephedrine markets and document a roughly 55% decline in methamphetamine use among arrestees following the shock, with no significant change in property or violent crime rates. These studies suggest that even large supply-side interventions may have limited or mixed success, particularly once behavioral responses and substitution are taken into account.

The papers most closely related to this one focus on drug consumer responses to two papers on connectivity of opioid markets and consumer responses to shocks. Moore and Schnepel (2021)] document the Australian consumer response to an early heroin shortage in 2001. Following a rapid reduction in heroin availability in the country they show reductions in overdoses suggesting a reduction in use. This effect is

short lived however and is soon offset by substitution into other drugs and increases in crime including heroin. One way this paper differs is in the lack of availability of heroin in the 2001 drought. There is no documented decrease in availability of the drug following the 2022 shock, indicating that effects are more likely to be driven by financial mechanisms. Deiana et al. (2024) also use price shocks to opium markets in Afghanistan in the early 2000's and trace decreases in opium prices to increases in the amount of opioids prescribed in U.S. counties. This documents how impactful Afghan opium markets across the world.

A third stand of the literature concerns the relationship between drug use and crime. Drug markets may affect crime through several channels: (i) pharmacological effects of substance; (ii) systemic violence associated with illegal distribution; and (iii) economic crime when users commit offenses to finance consumption. Existing evidence has documented correlations between drug use and property crime, but causal identification remains challenging due to confounding factors such as policing intensity, local economic conditions, and unobserved preferences (cite key papers here). By combining an exogenous international supply shock with granular measures of local consumption and crime, I contribute to the understanding that budgetary or economically motivated crime channels impact decisions to commit crimes.

Finally, I contribute to the growing literature that uses wastewater-based epidemiology (WBE) to measure drug consumption. Wastewater monitoring provides an anonymous, population-level measure of consumption based on concentrations of drug metabolites in sewage, aggregated over catchment areas. This approach provides researchers with reliable consumption estimates and overcomes many of the identification issues associated with research in illicit markets. While wastewater data has been used extensively in public health, epidemiology, and related STEM fields, the application in economics is only beginning. To the best of my knowledge, Daniele et al. (20XX)] is the first economics paper to use wastewater data directly as a measure of drug consumption. They study supply shocks in cocaine markets in Colombia and link these shocks to violence along supply routes and to consumption in Europe. Using data from the European Union Drug Agency (EUDA), they show that increases in cocaine production in Latin America are followed by increased consumption across European countries, with increased effects for countries more linked to Colombian markets.

However, the EUDA data are limited in both the range of drugs analyzed and in the frequency of testing compared to the Australian data I employ in this paper. The Australian wastewater monitoring program has been widely used in epidemiology but less frequently connected to economic behavior and market mechanisms. Tschärke et al. (2026)] examine the effects of an alcohol sales ban on alcohol consumption in Alice Springs, Northern Territory, using wastewater data to capture changes in population-level drinking. Wang et al. (2026)] use the same dataset to study changes in tobacco consumption in the context of a growing illegal tobacco market in Australia. They show identify an increased shift toward e-cigarettes that increasingly explain consumption in the illegal market. Verhagen et al. (2026)] focus on geographic determinants of drug use, showing that remoteness and community characteristics significantly influence cannabis consumption rates.

I am the first, to my knowledge, to use wastewater data to measure market-level impacts of a large,

international supply shock across a broad set of drugs and to combine these measures with detailed local crime data. I also use pre-shock wastewater consumption measures to control for geographic variation in baseline intensity of drug use across localities. This allows me to show that markets may differ systematically in their responses to a common shock depending on pre-existing consumption patterns and local characteristics. This idea is well established in the broader economics literature but has been absent in empirical work on illicit drug markets.

The remainder of the paper is organized as follows. Section 2 provides background on the Taliban’s opium ban and its impact on global opiate markets, along with an overview of the Australian drug market. Section 3 describes the wastewater monitoring program and the construction of my consumption measures. Section 4 presents a simple theoretical framework linking drug price changes to consumption and crime via budget constraints and substitution. Section 5 describes the data on wastewater, drug prices, and crime. Section 6 details the empirical strategy used to estimate price elasticities and crime effects. Section 7 presents the main results, and Section 8 concludes with a discussion of policy implications and directions for future research.

2 Background

2.1 Taliban Narcotics Ban

Following the 2020 peace agreement between the United States and the Taliban, the United States withdrew their forces from the country after twenty-three years of war. Following the withdrawal, the Taliban launched an offensive and quickly regained control of the country, becoming the de-facto government.

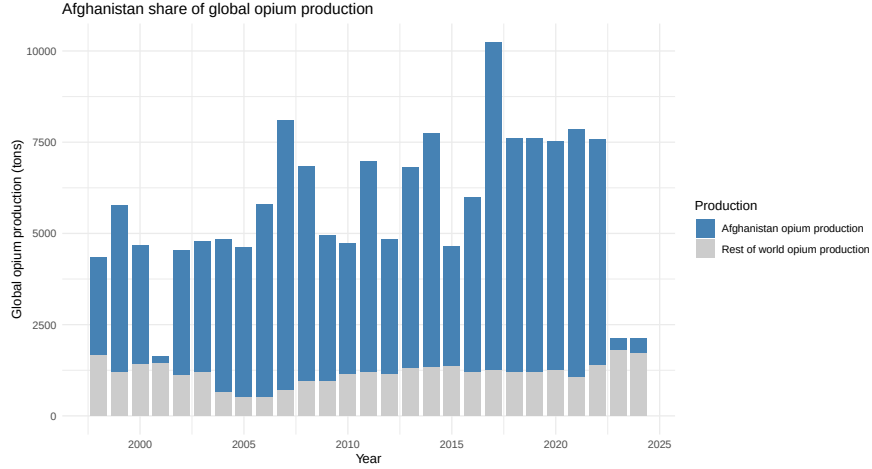
In April of 2022, the Taliban announced a ban on consumption and cultivation of narcotics within the country. The implications of this ban would be tremendous on the opium crop produced within the country. In 2021, Afghanistan accounted for 86% of global illicit opium production. At the time of the announcement, many were skeptical the ban would hold. For starters, revenue from the illicit opium market represented 9-14% of the country’s GDP in 2021. Many poor farmers relied on revenue from the crops to supplement their own income as well so enacting the ban would put the country economically at risk as well as increasing hardships for some of its poorest citizens. <https://globalinitiative.net/analysis/talibans-poppy-ban-afghanistan/>

In addition to the economic and political conditions within the country, many pointed to the fact that the Taliban had tried a similar ban in 2000 that was short-lived. The length of the ban may have been influenced by the U.S. invasion of the country in the following year and less due to economic conditions. Nonetheless in 2023, the year of the next growing season following the ban’s announcement, cultivation of opium in Afghanistan was down 95%. Now in its third year, the ban seems to be lasting with cultivation consistently down compared to pre-ban levels.

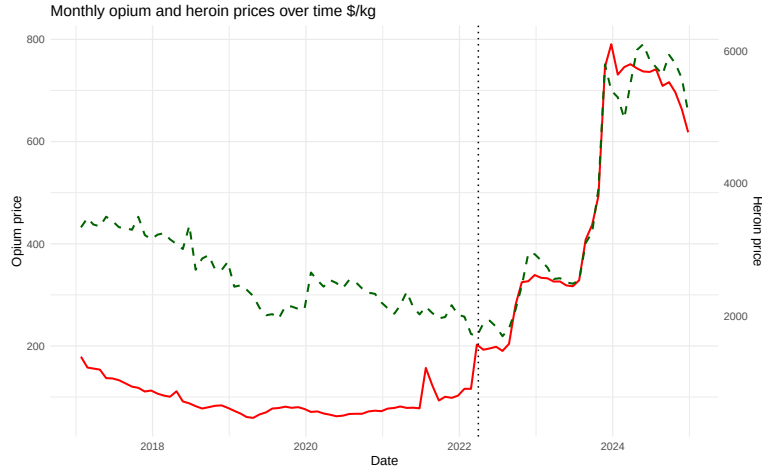
Figure 2 highlights the impact the ban had on global opiate markets. Panel (a) shows Afghanistan’s share of the global opium market from 1998-2024. Afghanistan has always been a vitally important country in the global opiate market accounting for the majority of production. Prior to the 2022 ban, 2001 also experienced

a similar drop in cultivation following the Taliban's first attempt at banning the product. Production is sizable in both 2000 and 2022, years the bans were announced, due to the timing of the announcements. The ban was announced in July of 2000 and April of 2022. Opium is typically harvested in the spring so the announcement was after the harvesting of the 2000 crop. In 2022 a two month grace-period was provided to farmers in order for them to harvest the current years crop. As a result both bans primarily impacted the following years sowing period.

In response to the shortage, some countries have increased production of illicit opium. Myanmar, now the world's largest illicit opium producer, increased domestic production by 35% from 2022 to 2023. Despite the increases in Myanmar and similar increases in production in other opium producing countries, the global supply is still down approximately 70% from pre-ban levels.



(a) Global Opium Production



(b) Opium and Heroin prices \$/kg

Figure 1: Afghan Opium Ban

Notes: Panels show impacts of the Taliban's opium ban on output and prices. All production and price data is from UNODC. Panel (a) shows global opium production by year from 1998-2024. Afghanistan's share of global production the blue (upper) section of the bar with all other countries production in grey (below). Panel (b) plots monthly wholesale prices for opium and heroin within Afghanistan. All prices are in USD and dotted line corresponds with the timing of the Taliban's announcement of the ban.

Panel (b) plots the monthly wholesale prices for farm-gate opium and heroin. Both are listed in USD per kilogram. Prior to the 2022 ban, prices remained relatively stable with a slight downward trend in the wholesale price of heroin. Faced with a fear of supply shortage, farm-gate prices of dried opium rose from \$202/kg in March 2022, to a high of \$790/kg in December of 2023. Wholesale prices for heroin similarly increased from \$1693/kg to \$5405/kg over the same time period. The price increases accelerated following the 2023 growing season, possibly as traders realized the ban efficacy in Afghanistan. Heroin prices, while smaller in absolute percentage changes, closely follow the patterns of opium as it is the primary ingredient in the production of heroin. There is a slight pre-trend in the two months prior to the announcement possibly indicating some level of insider-awareness of the coming ban although no reporting of this level of awareness

has been documented.

2.2 Drug Market in Australia

Australia’s drug market differs from other markets in developed countries in several ways. For one, Australia is geographically isolated. The country shares no land borders so all drugs produced outside the country need to enter through shipping or air cargo. Historically, Australia had a large heroin population operating within the country. As documented in Moore and Schnepel (2021)] several events transpired in 2001 to largely cutoff the nation from international supplies of heroin. This led to a rapid reduction in use across the country. Despite the reduction there has remained a small but persistent population of heroin users across the country.

A substantial amount of the drugs not produced within Australia are imported from the region known as the “golden triangle” which covers Thailand, Myanmar, and Laos. As this region represents some of the largest heroin producers in the world, the impact of the Taliban’s ban is likely more muted than say Europe, of which approximately 95% of heroin is sourced from Afghanistan. Still, this even further supports the fact that it is financial mechanisms and not lack of availability driving the effects shown in the paper.

The majority of the Australian population is located around the capital cities of each state which are all on or near coastal areas. One of the important aspects of the wastewater data used in this paper is the identification of variation in the intensity of consumption and drugs used across locations.

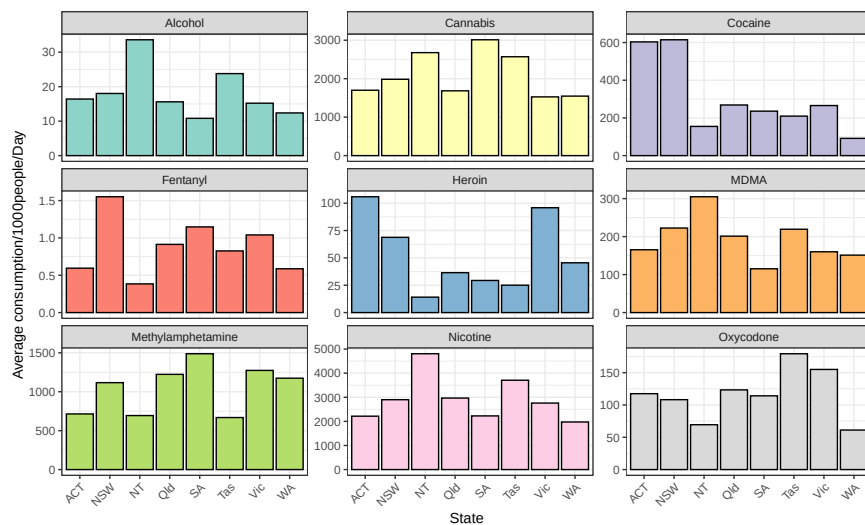


Figure 2: Average daily consumption by drug: Australian States and Territories

Notes: Figure 2 shows average consumption levels across Australian states and territories by drug type. Alcohol is reported in litres/1000people/day with all other drugs reported as mg/1000people/day. Due to differences in dosage, scales should not be compared across drug types.

Figure 2 shows average daily consumption of drugs across all Australian states and territories for 9 drug types as measured from the wastewater monitoring program. As we can see, states vary greatly by the types of drugs consumed. While Heroin has little use in northern territory, they have the largest consumption

levels for alcohol, nicotine and MDMA on a per-capita basis.

3 Tipping model for crime

Consider an individual who consumes a harmful good (e.g., an addictive substance) on a regular basis. The good is purchased out of available resources, which can be obtained either legally or illegally. When the price of the good increases, legal income alone may no longer suffice to finance the desired level of consumption, and the individual may then turn to illegal activities to supplement their budget.

Formally, consider an agent with an initial stock of resources B and an addiction level $s \in [0, 1]$. The agent chooses between a legal and an illegal source of income. If the agent works legally, they earn a wage w . If the agent engages in illegal activity, they obtain an illegal payoff b , but face a probability of capture $r \in [0, 1]$. If captured, they incur a punishment cost C , which can represent fines, imprisonment, loss of reputation and so forth.

The agent's utility from legal activity is given by

$$U_{\text{legal}} = V(w) - ps,$$

Where p is the price of the harmful good, and ps represents the, addiction-scaled, expenditure or disutility associated with financing consumption through legal means.

If the agent chooses the illegal option, expected utility is

$$U_{\text{illegal}} = (1 - r)[V(b) - c(s) - ps] + r[V(0) - C],$$

Where $c(s)$ is the, potentially addiction-dependent, non-pecuniary cost of committing the crime (e.g., moral cost).

The agent commits the crime if

$$U_{\text{illegal}} > U_{\text{legal}},$$

or explicitly,

$$(1 - r)[V(b) - c(s) - ps] + r[V(0) - C] > V(w) - ps.$$

Rearranging, the condition can be written as

$$rps > V(w) - (1 - r)[V(b) - c(s)] - r[V(0) - C].$$

Thus, the tipping price p^* above which the agent is willing to engage in crime is

$$p > p^*(s, r, C) = \frac{V(w) - (1 - r)[V(b) - c(s)] - r[V(0) - C]}{rs}. \quad (1)$$

Assuming that the legal and illegal payoffs (w, b) and the non-pecuniary crime $c(s)$ are held fixed, (1) implies that as $(s \rightarrow 1)$, the threshold price p^* at which the agent turns to crime generally falls. Individuals with higher addiction levels thus become willing to commit crime at lower price levels of the harmful good. This effect is reinforced if $c(s)$ is decreasing in s : higher addiction lowers the intrinsic disutility of crime, further reducing the tipping price. Similarly, for a fixed level of s increases in the market price will tip the marginal person into engaging in crime.

Now allow punishment levels and probabilities of capture to vary by crime type $j \in J$. Suppose each crime type j is characterized by an illegal payoff b_j , a capture probability r_j , and a punishment cost C_j . Then the tipping price for crime type j is

$$p_j^*(s, r_j, C_j) = \frac{V(w) - (1 - r_j)[V(b_j) - c(s)] - r_j[V(0) - C_j]}{r_j s}. \quad (2)$$

This expression makes clear that, for an individual on the margin of offending, crimes with lower expected sanctions—i.e., lower (r_j) and/or lower (C_j) , holding $(w, b_j, c(s))$ fixed—tend to have a lower tipping price p_j^* . Consequently, as the price of the harmful good rises and individuals become more likely to consider illegal means of financing consumption, they will optimally select crime types with relatively low effective expected punishment. In equilibrium, marginal offenders should therefore be disproportionately observed committing offenses with lower probabilities of capture or lower punishment severity.

We start from

$$p_j^*(s, r_j, C_j) = \frac{V(w) - (1 - r_j)[V(b_j) - c(s)] - r_j[V(0) - C_j]}{r_j s}.$$

Expanding the numerator,

$$\begin{aligned} p_j^*(s, r_j, C_j) &= \frac{V(w) - (V(b_j) - c(s)) + r_j(V(b_j) - c(s)) - r_j V(0) + r_j C_j}{r_j s} \\ &= \frac{V(w) - V(b_j) + c(s)}{r_j s} + \frac{V(b_j) - c(s) - V(0) + C_j}{s}. \end{aligned}$$

Equivalently,

$$p_j^*(s, r_j, C_j) = \underbrace{\frac{V(w) - V(b_j) + c(s)}{r_j s}}_{\text{depends on } r_j} + \underbrace{\frac{V(b_j) - c(s) - V(0) + C_j}{s}}_{\text{independent of } r_j}.$$

This makes the comparative statics transparent. First, the effect of the capture probability:

$$\frac{\partial p_j^*}{\partial r_j} = - \frac{V(w) - V(b_j) + c(s)}{r_j^2 s}.$$

Under the natural assumption that

$$V(w) - V(b_j) + c(s) > 0,$$

we have

$$\frac{\partial p_j^*}{\partial r_j} < 0.$$

Thus, a higher probability of capture r_j lowers the tipping price p_j^* ; equivalently, crimes with a lower probability of capture have a higher tipping price and remain attractive up to higher prices of the harmful good.

Next, consider the effect of punishment severity. From the expanded form,

$$p_j^*(s, r_j, C_j) = \frac{V(w) - V(b_j) + c(s)}{r_j s} + \frac{V(b_j) - c(s) - V(0)}{s} + \frac{C_j}{s},$$

so

$$\frac{\partial p_j^*}{\partial C_j} = \frac{1}{s} > 0.$$

Therefore, a higher punishment cost C_j raises the tipping price, whereas crimes with lower punishment severity C_j have lower tipping prices and are chosen at lower price levels of the harmful good.

4 Data

I use data from three main sources to identify the effects.

4.1 UNODC Drug Prices

The UN's Office of Drug Crime (UNODC) provides worldwide monitoring of drug prices throughout the world. Typically, countries law enforcement agencies submit prices they have identified in their own countries either through drug seizures or undercover purchases of drugs. In some cases, such as in Afghanistan, the UNODC will conduct special projects to monitor areas deemed problematic. In the case of Afghanistan, the UNODC conduct crop monitoring through a combination of satellite imaging and on the ground reporting to track changes in the heroin and opium market within the country.

Monthly farm-gate prices for opium and market prices for heroin are available at the wholesale level. All prices are reported in U.S. dollars per kilogram of the product.

4.2 National Wastewater Drug Surveillance Program

Wastewater monitoring initiatives in Australia began in 2009. In response to perceived escalating drug issues within the country, the Australian government implemented the National Wastewater Drug Surveillance Program in 2016¹. This program is notable for being the most comprehensive globally in terms of the variety of drugs measured, length of operation and the frequency of testing. The Australian Criminal Intelligence Commission (ACIC) administers the program, with analytical support from several Australian universities², ensuring coverage in all Australian states.

Approximately 50 wastewater treatment plants (WWTPs) participate in the program, representing around 50-60% of the Australian population. Each wastewater testing area corresponds to a specific sewage

¹Report 1 of the National Wastewater Drug Monitoring Program[?]

²University of Queensland and University of South Australia

catchment zone³. Although these catchment areas do not align strictly with defined neighborhood boundaries, they effectively delineate sections of urban environments. WWTPs may periodically enter or exit the program, resulting in minor fluctuations in the population under surveillance. Wastewater sampling is conducted in every state capital, along with several regional locations⁴. While the exact locations of the WWTPs are confidential, sites are coded as either capital city or regional location and their corresponding states are provided. Since each state contains one capital city, this facilitates the identification of capital city locations. Regional testing sites, being unidentifiable, are omitted from the analysis, ultimately including around 20 WWTPs in the study.

A crucial point is that while approximately half of the Australian population is monitored, nearly 100% of individuals residing within the designated geographic areas are included. For individuals within a sewage catchment area, inclusion in the sample hinges solely on their use facilities that feed into the wastewater system. Moreover, while regional locations are excluded from the main analysis, they represent a small fraction of the overall sample population. Figure 8 illustrates the population coverage for capital cities. According to the 2021 Australian Census, the cumulative population of the corresponding capital cities is approximately 13.3 million. Thus, although the program covers about half of the Australian population, around 85% of capital city residents are represented in the sample⁵.

Daily composite samples are collected by participating WWTPs over seven consecutive days. In certain instances, collecting samples for a full seven days may be impractical due to weather conditions or special events that could distort results. The primary objective of wastewater monitoring is to provide a reliable estimate of regular drug use within a specific area. For example, an event such as a music festival within a city’s catchment area could result in an influx of users of the municipal sewage system and an uptick in drug consumption, potentially skewing conclusions regarding drug use in the broader population. Additional details regarding the sampling protocol and relevant quality controls are discussed in Irvine et al. (2011)[6], Lai et al. (2011)[?], Lai et al. (2015), Tschärke et al. (2016)[7], and Bade et al. (2019)[1].

³Visual example of a catchment zone is listed in Appendix

⁴Regional locations are defined as non-capital cities, and are typically large population centers within each state.

⁵see figure 4 in appendix A for coverage details

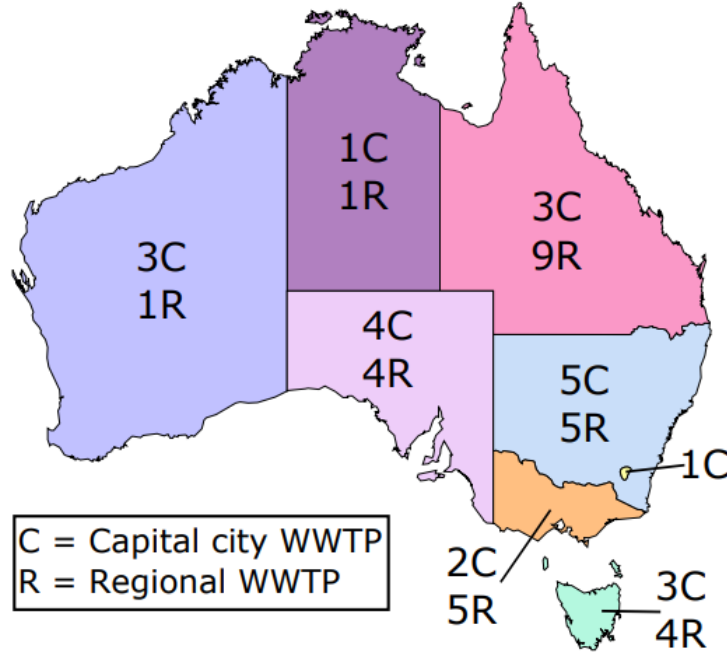


Figure 3: Testing site coverage

Notes: The figure shows the geographic coverage of the wastewater monitoring program in Australia. States and territories are divided by color and list the number of testing sites operating in each region. C indicates a site in a capital city and R a regional location. Regional is defined as non-capital city. Exact locations for each site is not known, but because capital cities are unique to each state data from these locations can be linked to demographic and crime data at the capital city level

The samples collected are normalized by population to estimate drug consumption per 1,000 individuals per day, reported tri-annually for capital cities and regional locations. As testing sites provide repeated sampling over time, trends within specific states will accurately reflect changes in consumption within the corresponding catchment areas over time. Given that the catchment areas remain constant in the study, fixed effects can be applied to account for characteristics unique to each area.

The unique aspect of this data is that it highlights geographic variation in consumption across both the intensity of consumption and type of drug consumed. Figure 3 presents average consumption of heroin across 8 capital cities in Australia. Since sampling is already population normalized to mg/1000people/day the results in the figure present relative intensity of consumption in communities across Australia. What immediately becomes apparent is the variation in consumption. Melbourne has by far the largest level of consumption at approximately XXXX mgs/1000people/day where Darwin, Northern Territory has essentially no consumption.

4.3 Crime Data

Data on reported crimes are collected from Australian State police jurisdictions. Uniform crime reporting data is not available in Australia, so data is collected individually from state police departments. All states and territories in Australia with the exception of Tasmania are included. The data reported by state police

record a monthly count of reported crimes by crime type. Reporting standards differ across police jurisdiction resulting in differences in granularity of crime type.

Due to differences in reporting standards, it is impractical to compare across states by crime type. Crimes are first classified as “financially-motivated” and “personally-motivated”. The determination for classification of “financially-motivated” crimes is if money is the main result of the crime. Under this criteria crimes such as burglary, robbery, theft, larceny etc.. similarly “personally-motivated” crimes are typically described as causing physical harm without financial reward. Crimes such as homicide, assault, and sexual crimes meet this criteria.

Since the effects of a price increase may impact crimes differently, crimes are further broken down into sub-types. Personally motivated crimes are separated into “Assault” and “Homicide”, while financially motivated crimes are separated into “Theft”, “Burglary”, and “Robbery”. Crimes have differences in both moral and societal stigma as well as punitive outcomes. While both are considered societal taboos, it is unlikely that a person committing theft, a non-violent crime, is held to the same standards as someone committing a robbery, which while it may be non-violent often uses the implication of violence to succeed.

5 Estimation

Estimation follows in two parts to identify two separate research questions. The first is identifying the impacts the increase in global heroin prices has on local consumption of drugs in Australia. In this estimation I aim to identify own price elasticity for heroin as well as cross price elasticities for other drugs to suggest substitution patters. In the second stage, I use the finding from the price elasticity estimation to motivate analysis on local crime rates. Given low price elasticities, increased prices should induce higher financially motivated crime at the local level.

5.1 Price Elasticities & Substitution

I am interested in identifying the demand responses for a variety of drugs in response to the price increase in global heroin prices following the announcement of the Taliban’s ban. An ideal experiment to use would be an equation of the form:

$$\ln(Q_{wt}^i) = \beta_{OLS}^{i,h} \ln(P_{w,t-j}^h) + \gamma^i X_{st} + \epsilon_{st}^i \quad (3)$$

Where Q is consumption of drug i as reported at wastewater testing site w , $P_{s,t-j}^h$ is the j - month lagged retail price of a pure dose of drug (i), and ϵ is the error term. Typically, researchers would rely on purity adjusted prices to take into account the stickiness of nominal prices in the retail market. It is common for nominal prices a retail customer faces to remain constant over time. What does fluctuate is the purity of the product they are offered, impacting the implied pure-weight price of a retail drug over time. Retail prices for heroin are unobservable in Australia. As such, I rely on prices of opium and heroin in Afghanistan as a proxy for the presumed purity adjusted price increases in Australia.

Classic problems of simultaneity in supply and demand confound interpretation for this estimation. Instead, I rely on the sudden supply shock following the Taliban’s announcement of the ban in April of 2022. Due to the international aspect of the ban’s impact, it is assumed all locations in Australia are impacted by this ban leaving no control sample. To overcome this issue, I employ a continuous difference-in-differences estimation method

$$\ln(Q_{wt}^i) = \beta_{IV}^i \ln(P_{s,t-j}^o) * Q_w^H + \gamma^i X_{st} + \delta_w + \tau_m + \epsilon_{wt}^i \quad (4)$$

relying on variation in pre-shock heroin consumption to measure the differential intensity of treatment across locations. Q_w^H is a time-invariant measure of pre-shock heroin usage as recorded in the wastewater data. Location fixed effects (δ_w) absorb all time invariant differences across catchment areas, essentially serving as neighborhood fixed effects, while month fixed effects (τ_m) account for seasonality in both the production and consumption of drugs. β then measures the differential price elasticities w.r.t opium prices for higher and lower consumption areas. The vector X is comprised of state level controls including the log of the local unemployment rate, logged not in labor force (NILF) rate, proportions of the population married and male.

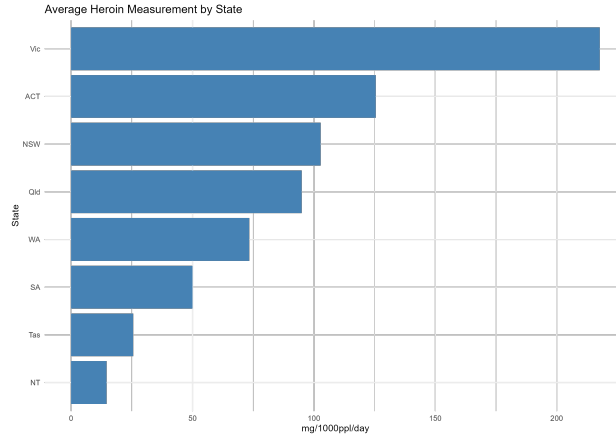


Figure 4: Average daily heroin consumption: Capital Cities

Notes: Figure 3 shows average consumption levels of heroin across all eight capital cities. Consumption is reported as mg/1000people/day.

To ease interpretation of the coefficient on the interaction, I transform the heroin consumption variable from daily milligrams consumed per 1000 people to daily doses consumed per 1000 people. The coefficient β can now be interpreted as the change in local crime for a change in the log price *and* a one dose increase in daily heroin consumption pre-ban. This serves as an appropriate proxy for the intensity of addiction in capital cities across Australia.

5.2 Crime Estimation

Due to the global impact of the Taliban’s ban, all heroin users within Australia are treated. To separate the impact of price increases on the propensity for crime, I employ a continuous differences-in-differences framework. Using the level of consumption in each Australian capital city prior to the ban creates a dosage framework for treatment. Some cities, like Sydney have nearly triple the concentration of heroin in the wastewater than cities like Darwin or Adelaide. Because the wastewater is already population normalized this can directly be translated into a per-capita level of heroin exposure in each city. I use the following form to estimate the effect of increased exposure to the heroin price increases, due to pre-ban heroin demand, on the local crime rate for crime j .

$$\ln(Crime_{ct}^j) = \beta_{CDD}^{i,o} \ln(P_{s,t-j}^o) * Q_c^h + \gamma^i X_{st} + \delta_w + \tau_m + \epsilon_{wt}^i \quad (5)$$

Where Q_c^h represents the pre-ban level of heroin in city c . I first run specifications for crimes broadly broken into personally and financially motivated crimes. These categories are determined by whether the outcome is deemed to have a financial or personal motivation as its outcome. Crimes such as theft, robbery, burglary, and prostitution would meet the definition for financially motivated crimes. Crimes such as assault, homicide, stalking and rape are deemed personally motivated.

I further separate crimes into subcategory to understand if specific crime types are driving trends. Indeed, allowing for heterogeneous probability of capture and/or punishment in equation (1) supports this as the tipping price increases with severity of punishment or probability of capture. I narrow analysis to five main groups ; “Assault”, “Homicide”, “Burglary”, “Robbery”, “Theft”. It would be reasonable to assume differential effects across these types. For instance, while assault and homicide are both considered personally motivated crimes, the punishment, investigative effort from police, and social taboo are much higher for homicide than assault suggesting a higher tipping price and thus muted impact of price increases on its probability of occurring.

6 Results

I examine how the sudden increase in opium prices stemming from the Taliban’s ban impacted drug markets and crime rates in Australia. The motivations for these are guided by several predictions. The first is that heroin is an inelastic drug. Due to the physical addictive-ness of the drug, users of the drug are likely less influenced by price movements and rather the prospects of facing withdrawal symptoms when weighing their consumption decisions. The second is that if substitutes are available, some users will switch consumption to other, possibly more harmful drugs. The third is that, given a low price elasticity, a user will be faced with an increased financial burden to maintain a given level of consumption. This increased financial burden may induce users to commit crimes in order to supplement their budgets.

6.1 Price elasticities

XXXX Preliminary Results XXXX

I begin by examining the impacts of the price increase on heroin consumption within Australia. Table 1 summarizes the effects across three specifications. Columns (1) and (2) show the raw effect of the global price of opium on consumption of heroin in Australia. The point estimates are statistically insignificant but both centered around zero. Columns (3-5) add the continuous treatment interaction of the price of opium over time with the pre-shock level of heroin consumption in each catchment area. Column (3) is run without any fixed effect. The coefficient on price is now significant at -0.1652 with no significance on the interaction term. Column (4) is run without controls but including month and catchment area fixed effects. This produces significant coefficients on both the lagged price trend and the interaction term. The interaction term has a positive coefficient indicating lower price elasticities in catchment areas with higher pre-shock heroin use. The last specification, column (5), includes both controls and fixed effects. For this specification, price produces a coefficient of -0.125, although it is no longer significant. The interaction term remains significant at a coefficient of 0.0009.

A large question in the literature is how interconnected are drug markets and to what degree do users switch between drugs. Table 2 presents the effects of the opium shock on the licit drug market. Column (1) measures the effects on nicotine use and column (2) shows the effect on alcohol use. Measurements in the wastewater differ slightly between these two. As with all other drugs shown, nicotine is reported in *mg/1000people/day* while alcohol is measures in *litres/1000people/day*. Labor market and demographic controls are included in both specifications in addition to month and catchment area fixed effects.

In both specifications the coefficients on price are negative and significant. The size of the magnitudes vary, with a coefficient on -0.0686 on nicotine and -0.1524 on alcohol. The interaction terms are positive and significant in both specifications as well. While the size of the effect is smaller than the specification in Table 1, the direction is the same. If nicotine and alcohol are complements to heroin, as the price trends would indicate, the mirrored directionality on the interaction terms helps to bolster the idea that areas with higher pre-shock levels of addiction exhibit lower price elasticities.

Table 3 presents results for illicit drugs. The same specification in Table 2 was run on a variety of illicit drugs. Column (1-5) present results for cannabis, cocaine, fentanyl, MDMA, methamphetamine, and oxycodone. Fentanyl and oxycodone are both opioids in the same drug class as heroin. While the Australian heroin market is smaller than that of Europe of the United States, substitution between these drugs can be problematic for policy makers. Indeed, the introduction of fentanyl into U.S. illicit opioid markets increased the overdose rate dramatically. While we see decreases in opioid consumption overall for these two categories, the interaction of price and pre-shock exposure produces positive and significant results for both drugs. Column (1) also presents positive results for the interaction on cannabis. Substitution from opioids into cannabis has been documented and has been suggested as a way for opioid users to reduce consumption.

Table 1: Heroin Consumption

	Log avg. consumption Heroin				
	(1)	(2)	(3)	(4)	(5)
Log opium price (t-3)	-0.0704 (0.0500)	-0.0199 (0.1261)	-0.1652** (0.0820)	-0.1410** (0.0653)	-0.1251 (0.1350)
Log unemployed		0.4271** (0.2116)			0.3369 (0.2136)
Log NILF		1.623 (1.302)			2.380* (1.372)
Log pop. 15–49		0.5666 (3.050)			0.6507 (3.115)
Constant			4.213*** (0.4084)		
Pre-shock heroin			0.0439 (0.0655)		
Log opium (t-3) $\times$ Pre			0.0178 (0.0132)	0.0176* (0.0104)	0.0189* (0.0104)
Month FEs	Yes	Yes	No	Yes	Yes
Catchment area FEs	Yes	Yes	No	Yes	Yes
Observations	327	327	318	318	318
R ²	0.62770	0.63756	0.34585	0.61676	0.62796
Within R ²	0.00652	0.03284		0.01613	0.04489

*Notes: Estimates report the effect of opium ban on heroin consumption, measured by monthly wastewater concentrations. Consumption is reported in mg/1000ppl/day. Column 1 has no FE or controls, column 2 includes month FE and catchment area FE to control for seasonality in drug consumption and production and location specific effects. Standard errors are clustered at the catchment area level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.*

Table 2: Licit Consumption

	Log avg. consumption	
	Nicotine	Alcohol
	(1)	(2)
Log opium price (t-3)	-0.0686* (0.0403)	-0.1524*** (0.0438)
Log unemployed	0.0502 (0.0639)	-0.0297 (0.0696)
Log NILF	-0.9481** (0.4094)	-1.740*** (0.4463)
Log pop. 15–49	1.889** (0.9269)	1.144 (1.014)
Log opium (t-3) $\times$ Pre	0.0088*** (0.0031)	0.0110*** (0.0033)
Month FEs	Yes	Yes
Catchment area FEs	Yes	Yes
Observations	316	316
R ²	0.71428	0.76060
Within R ²	0.05773	0.15591

*Notes: Estimates report the effect of opium ban on licit drug consumption, measured by monthly wastewater concentrations. Column 1 measures nicotine consumption in mg/1000ppl/day. Column 2 measures consumption in litres/1000ppl/day. Both specifications include month and catchment area FEs. Standard errors are clustered at the catchment area level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.*

XXXXXX Talk about if goods are substitutes or complements XXXX

6.2 Crime

XXXX Preliminary Results XXXX

I next move to estimating the effects of the price increase on local crime rates. The results from table 1 support the fact that heroin is inelastic. Given an inelastic good, price increases raise the financial strain on consumers. Given fixed budgets, consumers would need to find ways to supplement their incomes and, given the likely legal labor market constraints users of heroin face, user will switch to the illegal market. Table 4. presents the effects of the price-shock on crime rates in capital cities across Australia.

Because the exact locations of the catchment areas used for consumption are not known to me, I aggregate pre-shock heroin consumption to the capital city level. I run this specification for seven Australian capital cities, with Hobart, Tasmania left out due to lack of data. Crimes are grouped by broad crime type into financially motivated and personally motivated crimes. The classification of these crime types is determined by the ultimate result of the crime. Financially motivated results in monetary profit for the perpetrator and

Table 3: Illicit Consumption

	Log avg. consumption					
	Cannabis (1)	Cocaine (2)	Fentanyl (3)	MDMA (4)	Methamphetamine (5)	Oxycodone (6)
Log opium price (t-3)	0.0470 (0.0526)	-0.2412*** (0.0873)	-0.0660 (0.0973)	0.2263 (0.1372)	0.2983*** (0.0649)	-0.1384** (0.0539)
Log unemployed	-0.2552*** (0.0834)	0.3017** (0.1381)	0.3276** (0.1539)	1.380*** (0.2172)	0.1449 (0.1026)	0.1856** (0.0852)
Log NILF	1.589*** (0.5339)	-1.368 (0.8872)	-1.687* (0.9886)	-1.396 (1.395)	-0.8138 (0.6591)	-3.118*** (0.5476)
Log pop. 15-49	-1.538 (1.213)	11.61*** (2.014)	0.3029 (2.244)	-6.179* (3.167)	-4.922*** (1.496)	0.3749 (1.243)
Log opium (t-3) $\times$ Pre	0.0109*** (0.0040)	0.0055 (0.0067)	0.0146* (0.0075)	0.0101 (0.0105)	-0.0014 (0.0050)	0.0111*** (0.0041)
Month FEs	Yes	Yes	Yes	Yes	Yes	Yes
Catchment area FEs	Yes	Yes	Yes	Yes	Yes	Yes
Observations	316	318	318	317	318	318
R ²	0.84027	0.70583	0.63208	0.45117	0.66854	0.76937
Within R ²	0.14569	0.16127	0.05092	0.18266	0.09444	0.25954

Notes: Estimates report the effect of opium ban on illicit drug consumption, measured by monthly wastewater concentrations. All columns report consumption in mg/1000ppl/day. Both specifications include month and catchment area FEs. Standard errors are clustered at the catchment area level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

personally motivated is a crime on a person that does not results in monetary profit for the perpetrator. Crimes are normalized to the city population and reported as *crime/1000people/month*. Column (1) presents the results for financially motivated crimes. Similar to the coefficients in the elasticity specifications, the coefficients on price and the interaction term are significant with opposite signs. Price has a coefficient of -0.0775 indicating a general decrease in crime with respect to price with the positive interaction term indicating a muted effect in areas with higher pre-shock consumption of heroin.

Column (2) reports the effects on personally motivated crimes and primarily serves as a robustness check. If the mechanism for varying crime rates based on pre-shock heroin consumption is financially focused then we should not see increases in personally motivated crimes. Indeed the results show a negative, but statistically insignificant effect on price, but a negative and statistically significant coefficient on the interaction term.

The results in table 2 support the theory that financial burdens can induce crime as users seek to supplement their incomes through illegal means. We should not expect uniform effects across crime types however since the consequences, both legally and morally, vary by crime type.

Table 5 presents the same specification with results separating into subtypes. Columns (2,4,5) all report the results for financially motivated crimes. None of these specifications show significant results on the broad impact of opium prices on crimes of these types. All of these show positive and significant effects on the interaction. For a 1 dose increase in heroin consumption per 1000 people, burglary increases 0.0275, robbery by 0.0126, and theft by 0.0146.

Table 4: Financially vs. Personally Motivated Crimes

	Log financial crime rate Financially Motivated (1)	Log persons crime rate Personanlly Motivated (2)
Log opium price (t-3)	-0.0766** (0.0299)	-0.0301 (0.0228)
Log unemployed	-0.1596*** (0.0445)	-0.0451 (0.0339)
Log NILF	-0.0899 (0.2667)	-0.7608*** (0.2033)
log(TotalPop)	-13.77*** (3.868)	3.423 (2.949)
ProportionMarried	0.8348 (0.8068)	2.111*** (0.6152)
ProportionMale	20.69*** (7.564)	-0.9366 (5.768)
Log pop. 15–49	11.60*** (3.024)	0.5772 (2.306)
Pre-shock heroin $\times$ Log opium price (t-3)	0.0164*** (0.0028)	-0.0050** (0.0022)
Month FEs	Yes	Yes
State FEs	Yes	Yes
Observations	322	322
R ²	0.97661	0.98488
Within R ²	0.16715	0.37712

*Notes: Estimates report the effect of opium ban on prevalence of crime reported as crime/1000ppl/month. Data is limited to 7 Australian capital cities (excluding Hobart, Tasmania) Financially motivated includes all crimes with profit as the main outcome of the crime. Personally motivated includes crimes on a person, not included in Financially motivated. Both specifications include month and catchment area FEs. Standard errors are clustered at the city area level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.*

Surprisingly homicide also has a positive effect of 0.02. It is possible changes in the homicide rate are motivated by changes in the overall drug network within the country. Indeed, given increases in retail prices, drug turf may become more valuable leading to violence between sellers in the market. While descriptions of the homicides are not provided e.g. vehicular, gun violence, intimate partner, increased research into this effect is worthwhile.

Table 5: Crimes by Type

	Log_Count				
	Assault (1)	Burglary (2)	Homicide (3)	Robbery (4)	Theft (5)
Log opium price (t-3)	-0.0280 (0.0678)	-0.1031 (0.0694)	-0.0640 (0.0982)	0.0032 (0.0820)	-0.0802 (0.0567)
Log unemployed	-0.0526* (0.0248)	-0.0893 (0.1159)	0.1880 (0.2373)	0.0493 (0.1663)	-0.1082* (0.0523)
Log NILF	-0.7446* (0.3386)	0.2644 (0.6389)	0.1193 (1.114)	-1.093 (0.7421)	-0.6842 (0.5523)
log(TotalPop)	4.081 (10.60)	5.879 (5.318)	-17.54 (10.98)	-1.666 (5.819)	-9.478 (8.104)
ProportionMale	-0.3958 (15.32)	28.51** (10.18)	46.18 (27.55)	54.85*** (5.037)	18.46 (16.70)
ProportionMarried	1.648* (0.7407)	-0.0155 (1.162)	-4.296 (3.035)	0.1842 (2.292)	0.0386 (0.9221)
Log pop. 15–49	0.5966 (7.807)	-6.275 (4.980)	12.01 (9.630)	1.639 (5.178)	10.09 (6.132)
Pre-shock heroin $\times$ Log opium price (t-3)	-0.0047 (0.0069)	0.0275*** (0.0047)	0.0200** (0.0080)	0.0126*** (0.0033)	0.0145* (0.0072)
Month FEs	Yes	Yes	Yes	Yes	Yes
State FEs	Yes	Yes	Yes	Yes	Yes
Observations	335	335	275	335	335
R ²	0.99444	0.98544	0.77237	0.94728	0.99272
Within R ²	0.49464	0.13159	0.02644	0.10116	0.38941

*Notes: Estimates report the effect of opium ban on prevalence of crime reported as crime/1000ppl/month . Data is limited to 7 Australian capital cities (excluding Hobart, Tasmania). Columns report crime rates by subtype. Differences in police reporting require aggregation in some cities. Standard errors are clustered at the city area level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.*

7 Discussion & Conclusion

I provide new evidence of changes in consumption patters for drugs following a large market shock. I also document apparent spillovers from this shock into the persistence of crime in local areas. The shock used in this study is ideal as it mimics and extreme form of punitive supply reduction policies that seek to reduce access of drugs to the public. Several key findings stand out.

Looking first at drug consumption, following the increase in prices consumption for heroin does reduce. Despite this reduction, price elasticity for heroin seems to be lower than most other research suggests. I also

document that price elasticities appear to be lower in areas with higher pre-shock consumption highlighting that the level of addiction may influence responsiveness to market interventions. There appears to be fairly large rates of substitution into methamphetamine following the shock. Of special importance is substitution into other opioid drugs. Most notable is substitution into fentanyl which has a higher probability of overdose than heroin or oxycodone.

There are many reasons to suspect sellers would shift to sythetic drugs in the wake of the Taliban’s ban. Opium fields take time to develop and farmable land of this crop requires either explicit government permission or weak oversight. It would be hard to DTOs to quickly shift production into another country. Given the significant geographic shift in demand for raw opium, Myanmar is only able to increase production by 35%. Synthetic drugs, such as fentanyl, are relatively easier to produce and transport.

The strengths of wastewater monitoring programs can also be a weakness as drugs are uniquely identified by their metabolites. Fentanyl, which is in the drug class of nitazenes or synthetic opioids, is identified by the metabolite carafentanyl. There are many nitazenes and it is common for suppliers to introduce new ones into drug market, many of which are even more powerful than fentanyl. Other drugs in the nitazene drug class have different metabolites so would be unidentified in the wastewater data. Reports of overdoses in Australia cuased by nitazenes other than fentanyl have grown since monitoring began in 2021. It is then likely that the degree of substitution from heroin to other synthetic opioids is under measured in this paper.

Moving to crime, the results support the idea that financial pressure is a channel for crime prevalence. I identify that cities with higher levels of heroin consumption, prior to the Taliban’s ban, saw increased rates of financially motivated crime as the international price for opiates increased. In these same areas there also is a negative effect on personally motivated crimes. When separating crimes into subtype, burglary saw the largest impact of the ban, with significant results on robbery and theft. Surprisingly homicide also sees an increase. The mechanism for homicide may be different and shocks to supply markets may induce seller side fight for supply and territory.

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A Appendix A

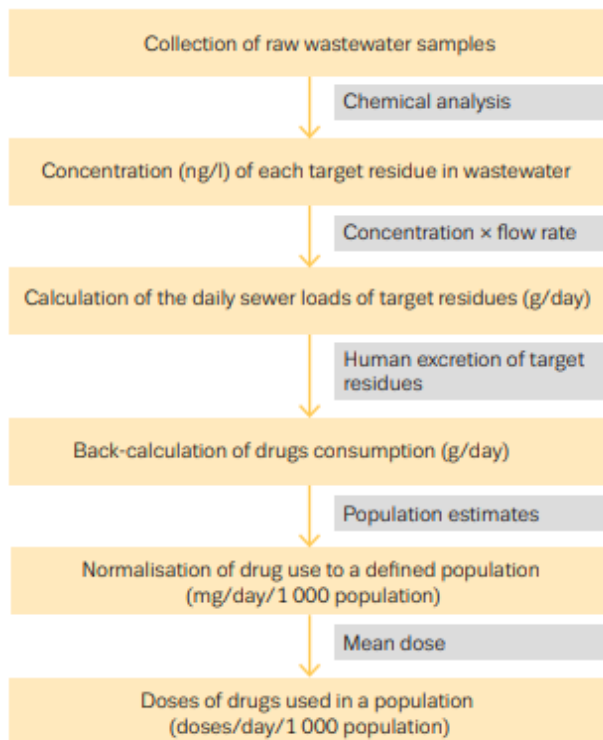
State or territory	April 2021 Capital	April 2021 Regional	June 2021 Capital
ACT	1	0	1
NSW	3	8	3
NT	1	1	1
Qld	3	8	3
SA	4	5	4
Tas	3	2	3
Vic	2	8	2
WA	3	4	3
Sites	20	36	20
Population (millions) C & R	11.2	1.8	11.2
% of Australian Population	47.9	7.9	47.9
Total population (millions)	13.0		11.2
% of Australian population	55.8		47.9

Estimates have been rounded to the nearest 0.1 million. Census 2016 population used (23,401,892) for population percentage estimates.

Figure 5: Population coverage: Wastewater Monitoring

FIGURE 1.1

The main consecutive steps of the wastewater analysis approach and the data required for each step



NB: Modified from Castiglioni et al., 2014.

Figure 6: Wastewater-based epidemiology workflow

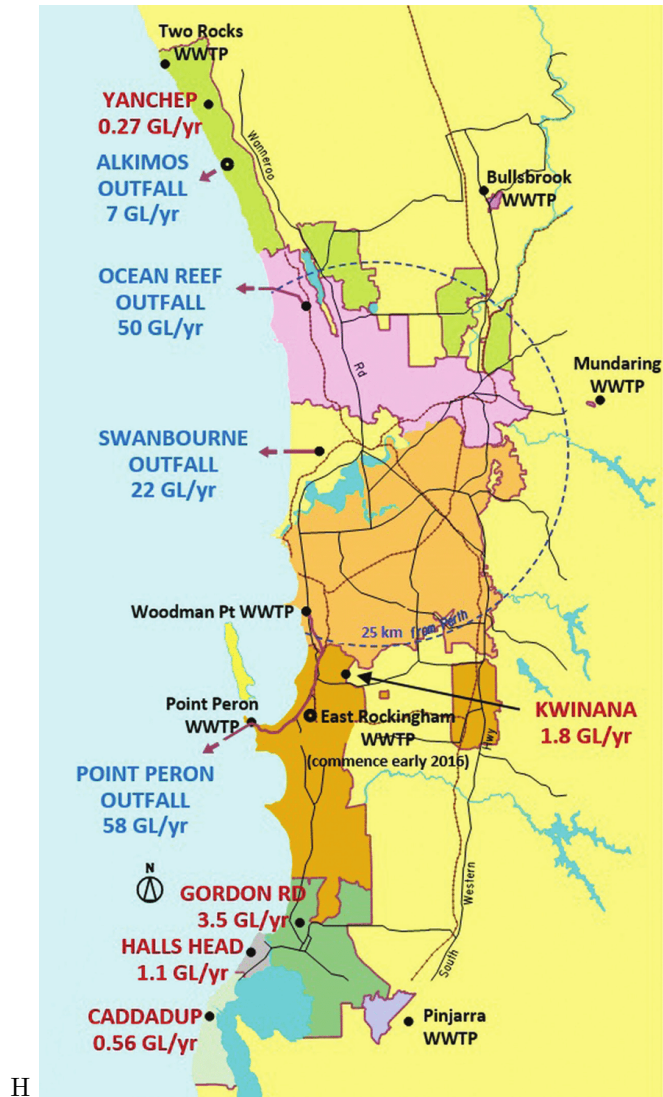


Figure 7: Perth, Western Australia catchment areas